UNIVERSITÀ DEGLI STUDI DI UDINE

Laboratorio di Programmazione Web
(laurea in Scienze e Tecnologie Multimediali)

A.A. 2026/2027

Web link alle pagine del corso:

http://elearning.uniud.it/moodle/course/view.php?id=6288

Server side programming
Database e MySQL/PHP

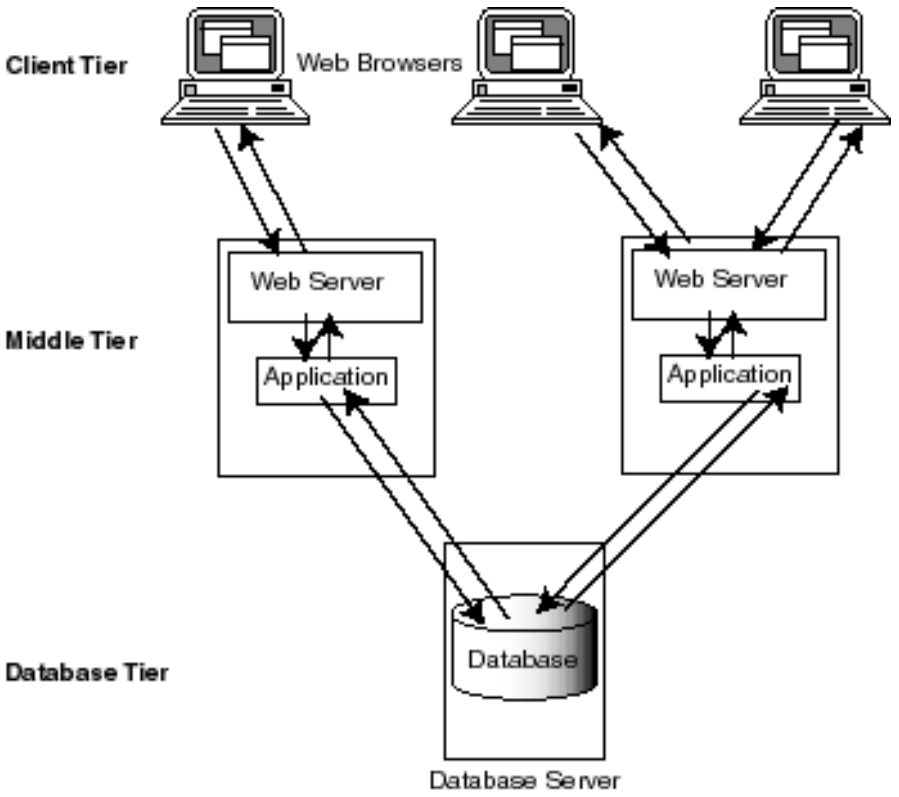
I materiali in queste slide sono tratti in parte da:

V. Della Mea, L. Di Gaspero, I. Scagnetto,"Programmazione lato server", Apogeo.
http://www.w3.org/
http://en.wikipedia.org (Wikipedia)
http://www.php.net

Database systems: Web applications

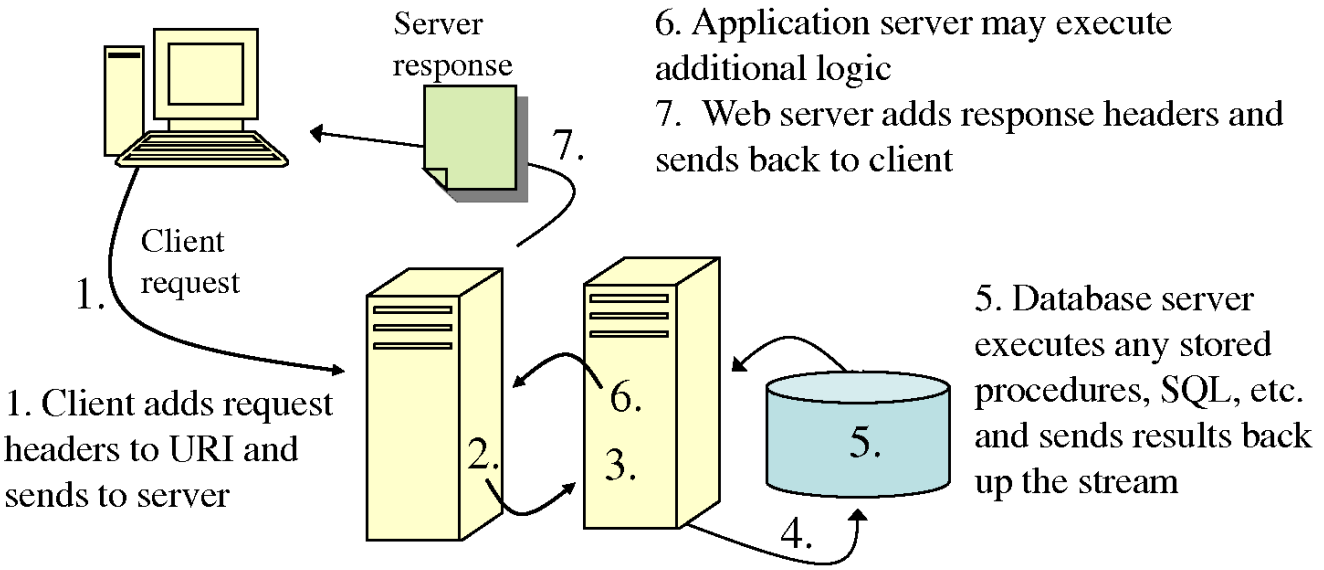
I. Introduction

Web server applications may access DataBase Management Systems (DBMS) to store/retrieve data to process



Database systems: Web applications

I. Introduction



2. Web server decodes and validates request
3. Application server executes business logic associated with resource
4. Connection made to database server

Database systems: Web applications

I. Introduction – Database and DBMS

- **Database:** a collection of inter-related data which is can be effectively retrieved, inserted and deleted. A database is used to organize the data in the form of a table, schema, views, and reports, etc

- **Database Management System (DBMS):** a software which is used to manage the database through an interface to perform various operations like database creation, storing data in it, updating data, creating a table in the database, and so on

I. Introduction – Database and DBMS

Key Features of DBMS

• **Data modeling:** A DBMS provides tools for creating and modifying data models, which define the structure and relationships of the data in a database

•**Data storage and retrieval:** A DBMS is responsible for storing and retrieving data from the database, and can provide various methods for searching and querying the data

•**Concurrency control:** A DBMS provides mechanisms for controlling concurrent access to the database, to ensure that multiple users can access the data without conflicting with each other

I. Introduction – Database and DBMS

Key Features of DBMS

•**Data integrity and security:** A DBMS provides tools for enforcing data integrity and security constraints, such as constraints on the values of data and data access control

•**Backup and recovery:** A DBMS provides mechanisms for backing up and recovering the data in the event of a system failure.

•**DBMS can be classified into two types:** Relational Database Management System (RDBMS) and Non-Relational Database Management System (NoSQL or Non-SQL)

I. Introduction – Database and DBMS

Key Features of DBMS: Relational vs other types

- **RDBMS:** Data is organized in the form of tables and each table has a set of rows and columns. The data are related to each other through primary and foreign keys.

- **NoSQL:** Data is organized in the form of key-value pairs, documents, graphs, objects, and so on. These are designed to handle large-scale, high-performance scenarios.

Database systems

II. Relational model

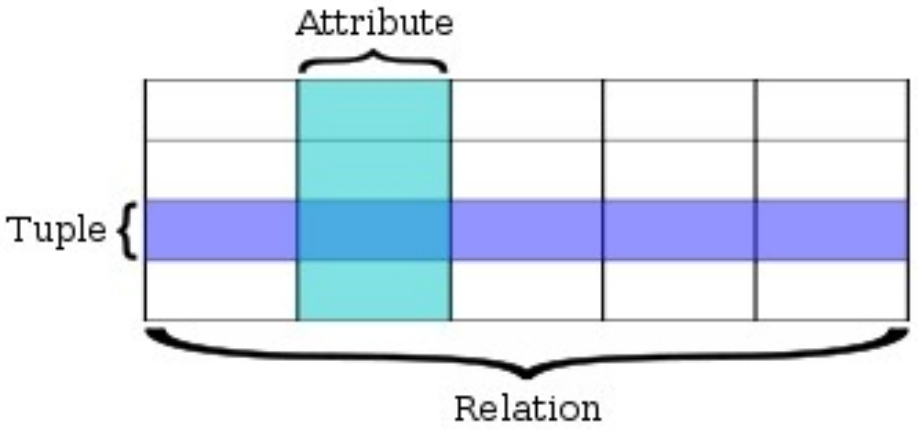
1. Data are organized as *Relations* (tables).

2. Tables are divided into columns, in which values of a given domain are stored.

3. Columns are named *Fields* or *Attributes*.

4. In each raw, named a *Tuple*, a data set with specific values in the range of each field, is stored

5. The set of attribute names is called the relation *Scheme*



II. Relational model

Example: *a flight time schedule*

1. Name of Relation: “ Voli ”

2. Scheme of the relation:

   Voli(Codice Compagnia, Numero Volo, Aerop. Partenza, Aerop. Arrivo, Ora partenza, Ora Arrivo)

3. Primary Keys: a set of attributes that uniquely specifies a tuple within a table

primary keys

Codice Compagnia	Numero Volo	Aeroporto Partenza	Aeroporto Arrivo	Ora Parten-za	Ora Arrivo
KL	1650	VCE	AMS	06:15	08:15
AZ	1358	TRS	FCO	11:05	12:15
AF	1427	VCE	CDG	12:40	14:30
LH	3999	TRS	MUC	11:15	12:25
AZ	324	FCO	CDG	14:40	16:50
BA	2583	VCE	LGW	12:25	13:40

Scheme

Tuple

II. Relational model

1. Foreign key: a field in a relational table that matches the primary key column of another table.

2. The foreign key can be used to cross-reference tables

Table: “Voli”

Codice Compagnia	Numero Volo	Aeroporto Partenza	Aeroporto Arrivo	Ora Partenza	Ora Arrivo
KL	1650	VCE	AMS	06:15	08:15
AZ	1358	TRS	FCO	11:05	12:15
AF	1427	VCE	CDG	12:40	14:30
LH	3999	TRS	MUC	11:15	12:25
AZ	324	FCO	CDG	14:40	16:50
BA	2583	VCE	LGW	12:25	13:40

Table: “Aeroporti”

Codice Aeroporto	Città	Paese
AMS	Amsterdam (Schiphol)	Paesi Bassi
FCO	Roma (Fiumicino)	Italia
TRS	Trieste (Ronchi dei Legionari)	Italia
VCE	Venezia (Tessera)	Italia
MUC	Monaco di Baviera	Germania
CDG	Parigi (Charles De Gaulle)	Francia
LGW	Londra (Gatwick)	Regno Unito
MXP	Milano (Malpensa)	Italia
BRU	Bruxelles (Zavantem)	Belgio

III. The SQL language for operations on databases

1. Database creation: **CREATE DATABASE** “database_name“

2. Table creation:

CREATE TABLE “table_name“ (
"column_1” “data type of column 1",
"column_2" "data type of column 2",
…
)

Example: creation of tables *“Voli” , “Aeroporti”*

```
CREATE TABLE Aeroporti (
        CodiceAeroporto CHAR(3),
        CittÃ  VARCHAR(255) NOT NULL,
        Paese VARCHAR(31) NOT NULL,
        PRIMARY KEY(CodiceAeroporto)
);

CREATE TABLE Voli (
        CodiceCompagnia CHAR(3),
        umeroVolo INTEGER,
        AeroportoPartenza CHAR(3) NOT NULL REFERENCES Aeroporti(CodiceAeroporto),
        AeroportoArrivo CHAR(3) NOT NULL REFERENCES Aeroporti(CodiceAeroporto),
        OraPartenza TIME NOT NULL,
        OraArrivo TIME NOT NULL,
        PRIMARY KEY(CodiceCompagnia, NumeroVolo)
);
```

III. The SQL language for operations on databases

Attributes data types

INTEGER	integer numbers (8-128 bit)
FLOAT	float numbers (32-64 bit)
DATE	date type (year-month-day)
TIME	time type (hh:mm:ss)
CHAR	fixed length strings
VARCHAR	variable length string

III. The SQL language for operations on databases

1. Table erase: **DROP TABLE** [IF **EXIST**] table_name

2. Select data from a database (query, extract data from table):

SELECT column_name, column_name, … **FROM** table_name

SELECT * **FROM** table_name

SELECT * **FROM** table_name **WHERE** condition(s)

Example: **SELECT * FROM** Voli **WHERE** CodiceCompagnia = 'AZ' **AND** OraPartenza > 12:00

Codice Compagnia	Numero Volo	Aeroporto Partenza	Aeroporto Arrivo	Ora Partenza	Ora Arrivo
KL	1650	VCE	AMS	06:15	08:15
AZ	1358	TRS	FCO	11:05	12:15
AF	1427	VCE	CDG	12:40	14:30
LH	3999	TRS	MUC	11:15	12:25
AZ	324	FCO	CDG	14:40	16:50
BA	2583	VCE	LGW	12:25	13:40

III. The SQL language for operations on databases

1. Raw delete: **DELETE FROM** table_name [**WHERE** condition(s)]

Example: **DELETE FROM** Voli **WHERE** Codice Compagnia = 'BA' **AND** NumeroVolo = 2583

Codice Compagnia	Numero Volo	Aeroporto Partenza	Aeroporto Arrivo	Ora Partenza	Ora Arrivo
KL	1650	VCE	AMS	06:15	08:15
AZ	1358	TRS	FCO	11:05	12:15
AF	1427	VCE	CDG	12:40	14:30
LH	3999	TRS	MUC	11:15	12:25
AZ	324	FCO	CDG	14:40	16:50
~~BA~~	~~2583~~	~~VCE~~	~~LGW~~	~~12:25~~	~~13:40~~

2. Raw insert: **INSERT INTO** table_name [column_name(s)] **VALUES** (value(s))

Example: **INSERT INTO** Aeroporti **VALUES** ('FCO','Roma (Fiumicino)','Italia')

Codice Aeroporto	Città	Paese
AMS	Amsterdam (Schiphol)	Paesi Bassi
FCO	Roma (Fiumicino)	Italia
TRS	Trieste (Ronchi dei Legionari)	Italia

III. The SQL language for operations on databases

1. Update raws: **UPDATE** table_name **SET** column_name = value [**WHERE** condition(s)]

Example: **UPDATE** Voli **SET** OraPartenza = 13:00 **WHERE** Codice Compagnia = 'BA' **AND** NumeroVolo = 2583

Codice Compagnia	Numero Volo	Aeroporto Partenza	Aeroporto Arrivo	Ora Partenza	Ora Arrivo
KL	1650	VCE	AMS	06:15	08:15
AZ	1358	TRS	FCO	11:05	12:15
AF	1427	VCE	CDG	12:40	14:30
LH	3999	TRS	MUC	11:15	12:25
AZ	324	FCO	CDG	14:40	16:50
BA	2583	VCE	LGW	12:25	13:40

13:00

The MySQL API for PHP

IV. MySQLi: PHP functions for database operations

1. Connect to database server

   resource mysqli_connect (**string** server_name, **string** username, **string** password ,[**string** db_name])

2. Select database on connected server

   bool mysqli_select_db (**string** db_name)

3. Perform query

   resource mysqli_query (**string** query)

4. Fill associative array with query result

   array mysql_fetch_array (**resource** query_result)

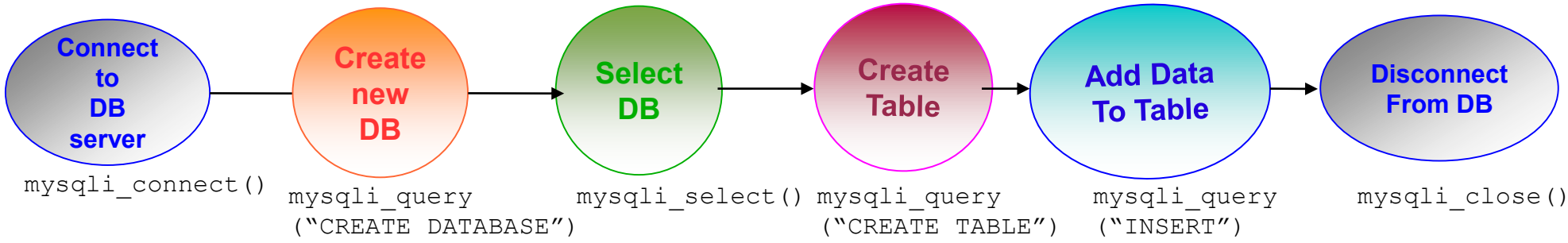
5. Disconnect from database server

   bool mysqli_close (**resource** connection_name)

The MySQL API for PHP

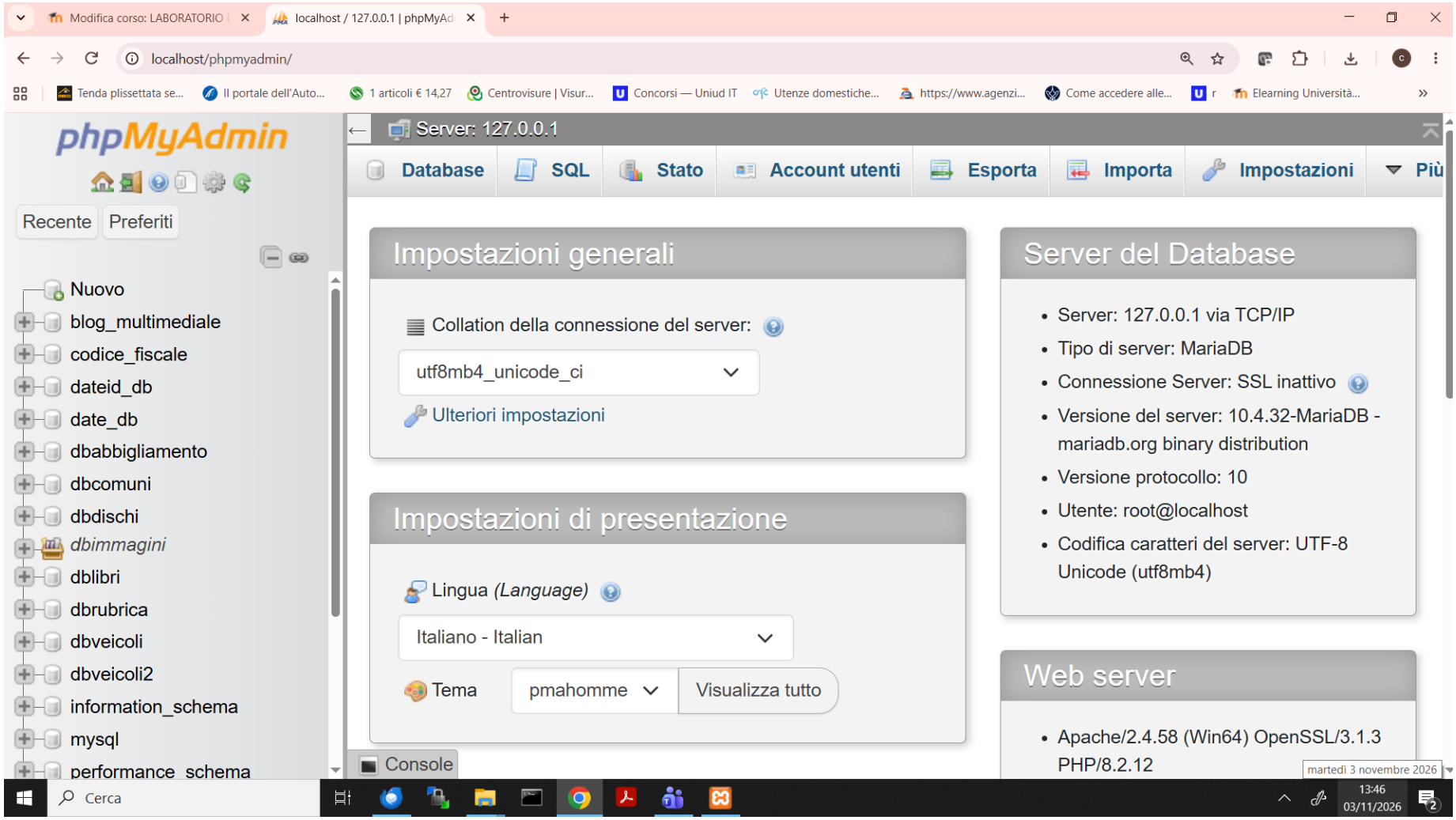
IV. MySQL/PHP: operations on database

Creating a database



XAMPP GUI for MySQL database

⟹ http://localhost/phpmyadmin/



The MySQL API for PHP

IV. PHP functions for database operations

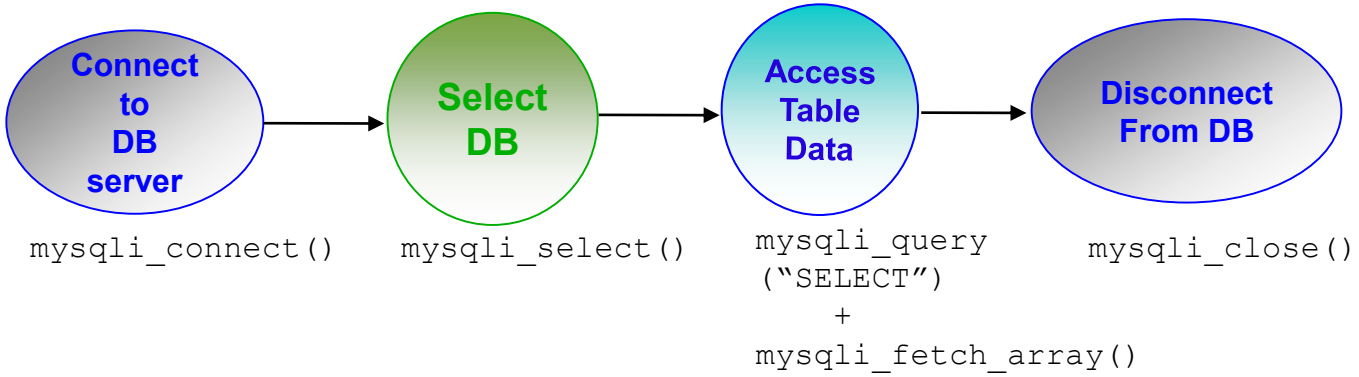
- ***Example:*** *create a new database and fill it with some new data*

```
1  <html>
2   <head><title>MySQL in PHP: Create a new database </title></head>
3   <body>
4    <h1>Create a new database</h1>
5    <?php
6     $host="localhost"; $username="root"; $password="";
7     $link = mysqli_connect($host,$username,$password);
8     $query="CREATE DATABASE person_db";
9     mysqli_query($link,$query);
10    mysqli_select_db($link,"person_db");
11    $query="CREATE TABLE Persons(Firstname CHAR(30),Lastname CHAR(30),Age INT)";
12    mysqli_query($link,$query);
13    $query="INSERT INTO Persons (FirstName, LastName, Age) VALUES ('Tim','Jones','31')";
14    mysqli_query($link,$query);
15    mysqli_close($link);
16   ?>
17  </body>
18 </html>
```

The MySQL API for PHP

IV. MySQL/PHP: operations on database

Accessing a database



The MySQL API for PHP

IV. MySQL/PHP: selecting a database

1. Since PHP ver. 8.xx, MySQL function `mysqli_select($link,'db_name')` will throw a **Fatal error** if dabase 'db_name' does not exist

2. Also, function `mysqli_connect($host,$username,$password,'db_name')` will throw a **Fatal error** if database 'db_name' does not exists

3. To check if a database already exists:

```
1    if(! mysqli_select_db($conn,"persons_db"))
2        die("Selezione del database non riuscita");
```

```
1   <?php
2   try{
3      mysqli_select_db($conn,"persons_db"));
4   }
5   catch(Exception $e){
6      echo "Exception error: " . $e->getMessage();
7      die("DB Select Error.");
8   }
9   ?>
```

The MySQL API for PHP

IV. PHP functions for database operations

- ***Example:*** *creating a database if it does not exists*

```php
<?php
try{
   mysqli_select_db($link,"persons_db"));
}
catch(Exception $e){
   echo "Exception error: " . $e->getMessage();
   // creating database if it does ot exists
   $query="CREATE DATABASE person_db";
   mysqli_query($link,$query);
   mysqli_select_db($link,"person_db");
   $query="CREATE TABLE Persons(Firstname CHAR(30),Lastname CHAR(30),Age INT)";
   mysqli_query($link,$query);
   mysqli_close($link);
}
?>
```

The MySQL API for PHP

IV. PHP functions for database operations

- ***Example:*** *access to a database through the PHP MySQL library*

```
<html>
 <head><title>Accesso a database</title></head>
 <body>
  <h1>Elenco persone</h1>
  <?php
   $conn=mysqli_connect("localhost", "nomeutente_db","password_db");
   if(!$conn) die("Connessione non riuscita");
   try {mysqli_select_db($conn,"persons_db");}catch(Exception $e){die("DB select error");}
   $query = "SELECT * FROM Persons";
   $risultato = mysqli_query($conn,$query);
   if (!$risultato) die("Query fallita");
   echo"<ul>\n";
   while($record = mysqli_fetch_array($risultato)) {
     echo"<li><strong>", $record["LastName"],"</strong>: ", $record["Age"], "</li>";
   }
   echo"</ul>\n";
   mysqli_close($conn);
  ?>
 </body>
</html>
```